

07 — Budget Justification

TriadicFrameworks: An Open-Science Substrate for Regime-Aware Modeling, Scanning, and Simulation

Submitted to NASA High Priority Open-Source Science (HPOSS)

1. Overview

This budget justification outlines the resources required to complete the TriadicFrameworks project during the 12-month performance period. The proposed budget supports open-source development, documentation, integration testing, and community dissemination. All requested funds directly contribute to the deliverables described in the Scientific/Technical Plan.

The total requested budget is approximately \$100,000, consistent with typical HPOSS award levels.

2. Personnel

2.1 Principal Investigator (PI)

Effort: Part-time (approximately 25% FTE for 12 months)

Responsibilities include:

- scientific and technical direction,
- protocol development (dsrsp/0.1, RSM, vST, SLRP),
- documentation oversight,
- integration testing,
- community engagement,
- reporting and deliverable coordination.

Requested support covers a portion of the PI's time dedicated to development, documentation, and project management.

2.2 Contract Developers (1–2 part-time contributors)

Effort: Equivalent to ~1 full-time developer for 12 months (distributed across contractors as needed)

Responsibilities include:

- implementing dsrsp/0.1 reference libraries,
- developing RSM and vST alignment modules,
- building ILP module reference implementation,
- producing example datasets and notebooks,
- assisting with documentation and integration guides.

Developer support is essential for producing high-quality, open-source implementations and ensuring timely delivery of all technical components.

3. Other Direct Costs

3.1 Cloud Services and Hosting

Modest cloud resources are required for:

- repository hosting and continuous integration,
- documentation builds,
- storage of example datasets,
- testing of simulation-based workflows.

These costs are minimal but necessary for maintaining open, reproducible infrastructure.

3.2 Software and Tools

All development will use open-source tools whenever possible.

Budgeted costs cover:

- domain-specific libraries or simulation tools if required,
- minimal paid services for documentation or testing infrastructure.

3.3 Dissemination and Outreach

Funds support:

- hosting a virtual workshop or tutorial,
- preparing training materials,
- producing onboarding documentation for community contributors.

These activities directly support NASA's open-science goals.

4. Travel (Optional and Minimal)

If appropriate, limited travel funds may be allocated for:

- attending an open-science workshop,
- presenting project outcomes at a NASA-aligned event.

Travel is not required for project completion and will be minimized.

5. Indirect Costs

If applicable, indirect costs will follow the submitting organization's established rate. For small research initiatives or independent organizations, indirect costs may be minimal or waived entirely.

6. Budget Alignment with Deliverables

The requested budget directly supports the project's core deliverables:

- **dsrsp/0.1 specification and schemas**
(PI + developer time)
- **RSM and vST alignment layer**
(developer time + PI oversight)
- **reference implementations and ILP module**
(developer time)
- **integration and testing workflows**
(developer time + cloud resources)
- **documentation and integration guides**
(PI + developer time)
- **community workshop and dissemination**
(outreach costs)
- **archival with DOIs**
(minimal overhead)

Every cost is tied to a specific, measurable output.

7. Summary

The proposed budget is modest, focused, and aligned with HPOSS expectations. It supports the personnel, tools, and infrastructure necessary to deliver a fully documented, openly licensed scientific substrate that advances NASA's open-science mission. All requested funds directly contribute to the creation, testing, documentation, and dissemination of TriadicFrameworks components.